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Alzheimer's disease detection using deep learning and machine learning: a review

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Abstract
Alzheimer's disease (AD) is a progressive neurodegenerative disorder that significantly impacts cognitive function, posing challenges in early diagnosis and treatment. Advances in artificial intelligence (AI) have revolutionized medical image analysis, providing robust frameworks for accurate and automated AD detection. This paper reviews recent developments in deep learning (DL) and machine learning (ML) models for AD classification, like convolutional neural networks (CNNs), transfer learning, hybrid architectures, and novel attention mechanisms. Additionally, applications of AD based on AI models, datasets, preprocessing techniques, challenges, and recent studies in this field are discussed. Also, the paper provides different medical modalities, factors of increasing risk of Alzheimer, progress stages of this disease, and several metrics of assessing AI models' performance. These metrics such as accuracy, matthews correlation coefficient (MCC), F1-score, recall, precision, area under the receiver operating characteristic (ROC) curve, confusion matrix, and loss. Further, the paper presents several comparisons of different DL approaches for AD, limitations, new trends, suggestions, and future directions for this evolving field.

Keywords Deep learning · Alzheimer disease · Artificial intelligence · Datasets · Transfer learning · Evaluation metrics · CNN · Medical applications

1 Introduction

Early detection of Alzheimer's disease (AD) is essential for implementing effective preventative strategies. As the most prevalent chronic condition among the elderly, AD affects a significant portion of the aging population, making timely diagnosis crucial to managing its high incidence rate. AD is a leading cause of dementia worldwide, necessitating timely

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and precise diagnostic techniques. So, there are multiple potential causes of AD that affect thinking, memory, and behavior in older individuals (Chaddad et al. 2018). Traditional approaches rely on clinical assessments and neuroimaging analysis, often involving significant manual intervention (Iqbal et al. 2023).

Deep learning (DL), with its ability to automatically extract meaningful features from data, has emerged as a transformative tool in medical imaging, particularly in detecting and classifying AD from modalities such as positron emission tomography (PET), magnetic resonance imaging (MRI), and computed tomography (CT) scans (Kaya and Çetın-Kaya 2024). Classification tasks in AD detection aim to categorize patients into healthy, mild cognitive impairment (MCI), or AD groups. This process helps in early intervention, particularly in MCI cases where the disease might progress to AD (Pei et al. Oct. 2022).

The progress of brain stages for AD is demonstrated in Fig. 1, which highlights the differences between No Impairment “normal cognition (NC)” as a healthy stage, mild decline “mild cognitive impairment (MCI)” as a midle stage, and severe decline “Alzheimer’s disease (AD)” as a clinical stage (Bellio 2021).

The likelihood of developing AD grows with age, particularly among individuals over 65 (Grundman and Petersen 2004). Several factors may contribute to an increased risk of Alzheimer, including: Genetics: specific genetic variations have been linked to a heightened risk of Alzheimer’s. Environmental Influences: exposure to harmful substances or experiencing head injuries can raise the likelihood of developing the disease. Lifestyle Choices: unhealthy habits such as poor nutrition, inactivity, and other detrimental lifestyle behaviors may play a role in increasing the risk. Medical Conditions: health issues like hypertension, diabetes, and elevated cholesterol levels are also associated with a greater risk of Alzheimer’s (Malik et al. 2024). Figure 2 demonstrates some factors of increasing risk of Alzheimer disease.

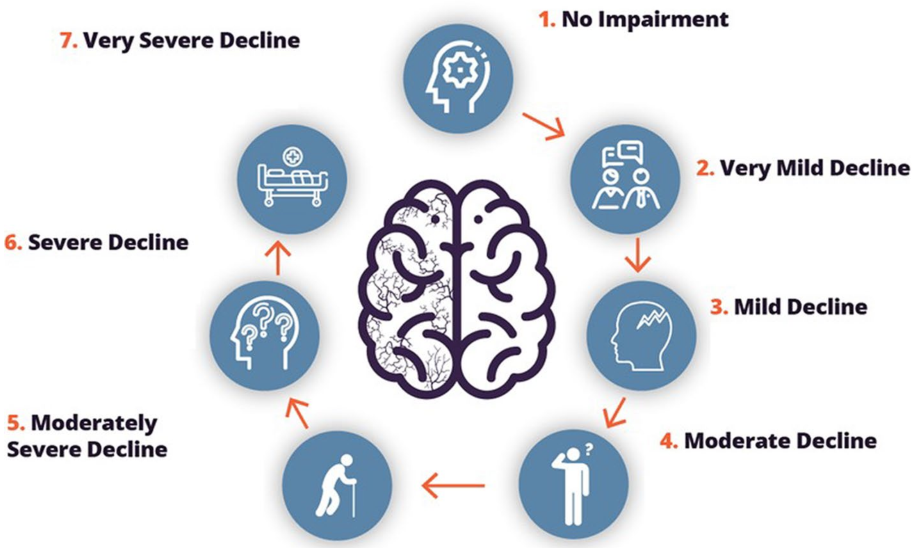

Fig. 1 Progress of brain stages for Alzheimer’s disease (Bellio 2021)

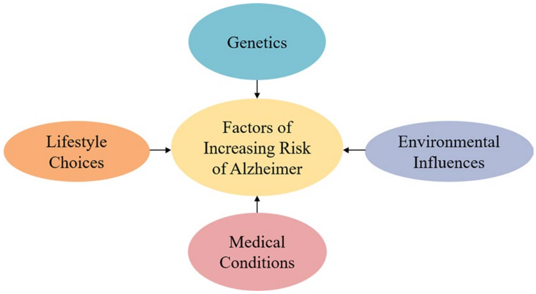

Fig. 2 Factors of increasing risk of Alzheimer disease

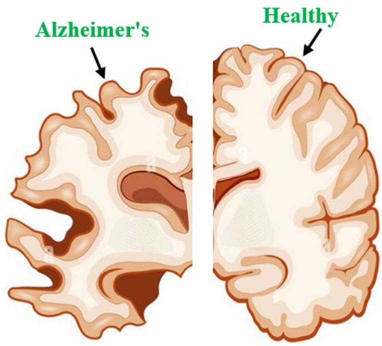

Fig. 3 Human brain mass for healthy and Alzheimer's

The commonly used neuroimaging modalities include MRI that provides structural information, enabling the detection of brain atrophy, PET that offers functional insights, capturing metabolic changes, and CT that helps in identifying structural abnormalities (Wu et al. 2019; AlSaeed and Omar 2022). Alzheimer's disease leads to a significant reduction in brain mass, Fig. 3 illustrates human brain mass for healthy and Alzheimer. In individuals with normal cognition, brain shrinkage is minimal or negligible. However, those with MCI experience an accelerated reduction in brain volume, losing around 1–2% annually—significantly faster than typical aging (Huang et al. 2020). In AD, the rate of brain volume loss increases further, reaching 3–5% per year. Regions like the hippocampus are particularly affected, with shrinkage rates as high as 10–15% annually in advanced stages (Sungura et al. 2021).

The main contributions of this review are the following:

- Summarizing the current AI models whether DL or ML for classifying Alzheimer's disease (AD) in brain MRIs.
- Reviewing several applications of AD detection using different AI models.
- Summarizing the datasets, preprocessing techniques, and various evaluation metrics of AI models for AD detection.
- Reviewing the recent studies in this field with comarisons of DL approaches for AD detection.
- Comparing AI Approaches versus traditional AD diagnostic methods.
- Presenting a discussion on real-world implementation of AI models and dataset adaptability.
- Providing a discussion on the overfitting problem in DL models and their mitigation

methods.
- Discussing the potential bias in AI models and mitigation strategies for this bias.
- Presenting a discussion on the challenges and limitations in this field of AD.
- Discussing the computational cost and latency in clinical AI deployment.
- Presenting new trends, suggestions, and future directions in DL for AD detection.
- Comparing several studies in AD detection in terms of the performance, methodology, and key contributions.

The remainder of this paper is organized as follows: Sect. 2 presents AI models for AD classification and detection. Section 3 provides several applications of deep learning in Alzheimer disease detection. Section 4 presents different dataset and preprocessing for Alzheimer disease detection. Section 5 presents recent studies in AD. Section 6 presents comparison of AI approaches versus traditional AD diagnostic methods. Section 7 discusses real-world implementation of AI models and dataset adaptability. Section 8 explains the overfitting problem in DL models and their mitigation methods. Section 9 provides some medical modalities used with AD. Section 10 presents the evaluation metrics used in AD detection, while the potential bias in AI models and mitigation strategies are presented in Sect. 11, while challenges and limitation in this field and comparisons of DL approaches for AD are introduced in Sect. 12. Section 13 presents a discussion about computational cost and latency in clinical AI deployment. New trends and suggestions in DL for AD detection are presented in Sect. 14. Future directions in deep learning for AD detection are discussed in Sect. 15. Section 16 introduces a benchmark comparison of DL models for AD detection. Finally, the paper is concluded in Sect. 17.

2 Key AI models for AD classification and detection

This paper reviews state-of-the-art artificial intelligence (AI) models employed for AD detection, highlighting classification methodologies, their performance, and the challenges faced in clinical translation. So, several advancements in AI models for AD classification are presented as follows.

Convolutional Neural Networks (CNNs) are widely used for image-based AD detection due to their ability to extract hierarchical features efficiently (Feng et al. 2019). They consist of convolutional layers that learn spatial patterns and pooling layers that reduce computational complexity while preserving essential information. CNNs have proven effective in tasks like brain tumor classification (Mohsen et al. 2023a), object detection (Zou et al. 2023), and segmentation (Sahu et al. 2018; Mohsen et al. 2024).

Long Short-Term Memory (LSTM) networks are a type of RNN designed for sequential data processing, addressing RNN limitations with memory cells and gating mechanisms. The forget gate removes irrelevant data, while input and output gates regulate information storage and usage (Nagabushanam et al. 2020; Chakraborty et al. 2020). Their ability to capture long-term dependencies makes them effective in tasks like speech recognition, machine translation, and stock price prediction.

Hybrid models integrate multiple deep learning paradigms, such as CNNs with attention mechanisms or RNNs, to enhance interpretability and performance (Mohsen et al. 2021a). They are valuable for complex tasks like medical imaging and disease diagnosis, where balancing accuracy and explainability is crucial. Examples include CNN + RNN models for analyzing time-series MRI data to predict disease progression and multi-branch hybrid models that combine CNNs trained on different modalities (e.g., MRI, PET) with a shared attention mechanism for comprehensive insights (Prakash et al. 2023).

Transfer learning are pre-trained DL models trained on huge datasets and designed to be fine-tuned for particular tasks. These models, such as BERT for language tasks or VGG16 for image tasks, leverage transfer learning to adapt general knowledge to domain-specific problems (Minoofam et al. 2023). Pre-trained models save time and computational resources while achieving high performance in tasks like medical imaging, sentiment analysis, and natural language processing. Also, pre-trained models such as Inception and ResNet have been adapted for AD detection. Fine-tuning these networks on medical imaging datasets reduces the need for extensive labeled data and accelerates model convergence. For example, combining CNNs with pre-trained VGG16 networks significantly improved classification accuracy in Alzheimer's studies.

Generative adversarial network (GAN) includes two networks—a generator and a discriminator—working in opposition. The generator creates fake data, while the discriminator evaluates whether the data is fake or real. Through this adversarial training process, the generator learns to produce highly realistic data. GANs are utilized in several tasks for example, image synthesis, deepfake creation, and data augmentation for training other models (Tiwari et al. 2020).

Autoencoder is a neural network designed to learn efficient representations of data through compression and reconstruction. It has two main components: an encoder that compresses the input into a latent representation and a decoder that reconstructs the input from this representation. Autoencoders are useful for tasks such as noise reduction, anomaly detection, and generating new data. Variational Autoencoders (VAEs) and Convolutional Autoencoders are common variants used in different domains (Failed 2019).

Vision transformer (ViT) introduces a transformer-based architecture to image processing, replacing the traditional convolutional approach. It divides images into patches, which are treated as input tokens for a transformer model. By applying self-attention mechanisms, ViTs focus on relationships between patches, making them effective for tasks like object detection, image classification, and segmentation. Their scalability allows them to excel with large datasets, though they may require significant computational resources (Gong et al. 2024).

Deep neural network (DNN) is a general framework for neural networks with multiple layers. It includes an input layer, hidden layers, and an output layer, with each neuron in one layer linked to each neuron in the next. DNNs use optimization algorithms like gradient descent and backpropagation to learn patterns from data. This versatile architecture is applied across diverse domains, including image recognition, and financial modeling, offering a robust foundation for many deep learning solutions (Wu et al. 2024).

Attention-based models, such as dual-attention mechanisms, focus on clinically relevant regions, improving detection accuracy and reducing false positives. The common recent attention models is based on a block, namely sequeeze and excitation (SE). Examples include self-attention in transformers or spatial and channel attention in convolutional settings. Attention models can highlight critical regions in brain MRI scans that contribute to disease diagnosis, aiding clinicians in understanding model decisions (Li et al. 2024).

K-nearest neighbors (KNN) is a simple, non-parametric supervised learning (SL) algorithm utilized for regression and classification applications. KNN works by recognizing the k closest data points to a provided query point, depend on a distance measurement like Euclidean distance, and then predicting the output based on the majority class in case classification or the mean for a regression task of these neighbors. It is easy to implement and interpretable but can be computationally expensive for huge dataset because of the need to calculate distances for all points. KNN performs well with small datasets and when features are scaled appropriately (Mohsen et al. 2021b).

Support vector machine (SVM) is a SL algorithm primarily utilized for classification applications, though it can also applied for regression products. The SVM operates via determining a hyperplane that optimum separates data points of various categories in a high-dimensional space. It maximizes the margin between the hyperplane and the nearest data points from each class, called support vectors. The SVM can also handle non-linear problems using kernel tricks like the radial basis function (RBF). It is effective in high dimensional spaces, but can be computationally intensive with large datasets (Kazemi et al. 2022).

Decision tree (DT) is a tree-structured model utilized for regression and classification tasks. It splits the dataset into subsets based on feature values, forming a tree with decision nodes and leaf nodes. Each decision node represents a test on a feature, while leaf nodes provide the output prediction (Rahmatillah et al. 2023). DT is easy to interpret both numerics and categorical data. However, it is prone to over-fitting, which can be mitigated using methods such as combining or pruning multiple trees in ensemble methods (Aaboub et al. 2023).

Ensemble learning is a powerful ML approach that combines multiple models, it improves model accuracy by combining multiple weak learners. Techniques include bagging (reducing variance, e.g., Random Forest), boosting (reducing bias, e.g., AdaBoost, Gradient Boosting), and stacking (using a meta-learner for better predictions). These methods often outperform individual models and are widely used in competitions like Kaggle for their robustness and ability to reduce overfitting. (Ramteke and Maidamwar 2023).

Artificial neural network (ANN) is a model inspired by biological neural networks (BNNs). It includes an input layer, one hidden layer, and an output layer. Each layer contains nodes linked by weights, which are updated during training to minimize the error. ANNs can model complex relationships in data and are utilized for applications like regression, classification, and pattern identification. However, they need significant computational resources and large datasets to perform well (Khan et al. 2023; Aswin et al. 2022).

Logistic regression is a model utilized for binary classification tasks. Despite its name, it is a classification algorithm rather than a regression method. Logistic regression uses the sigmoid function to map linear combinations of input features into a probability between 0 and 1. The output is thresholded (e.g., at 0.5) to assign class labels. Logistic Regression is simple and works well for linearly separable data, but it struggles with non-linear relations unless extended with feature transformations (Yadhukrishnan and MA 2023).

Naive bayes (NB) is a model based on Bayes' rule, assuming conditional independence between features given the class label. Despite its "naive" assumption of feature independence, it often performs surprisingly well for text classification and spam filtering. Variants include Gaussian NB (for continuous features) and Multinomial Naive Bayes (for count-based features like word frequency). It is efficient, requires minimal data, and is robust to irrelevant features, but its independence assumption may limit performance in certain datasets (Vijay and Verma 2023). Figure 4 summarizes most common AI models for AD classification.

3 Several applications of Alzheimer disease detection with AI models

Artificial Intelligence (AI) has revolutionized the field of medical imaging and diagnostic analysis. Its application in Alzheimer's disease (AD) detection spans a wide range of clinical, research, and technological domains. The applications of AI in this field span diagnostic, therapeutic, and research domains. By leveraging advanced AI architectures and integrating multimodal data, these models hold immense potential to improve early detection, streamline clinical workflows, and enhance personalized care for AD patients. We present many key applications with highlighting its impact as follows:

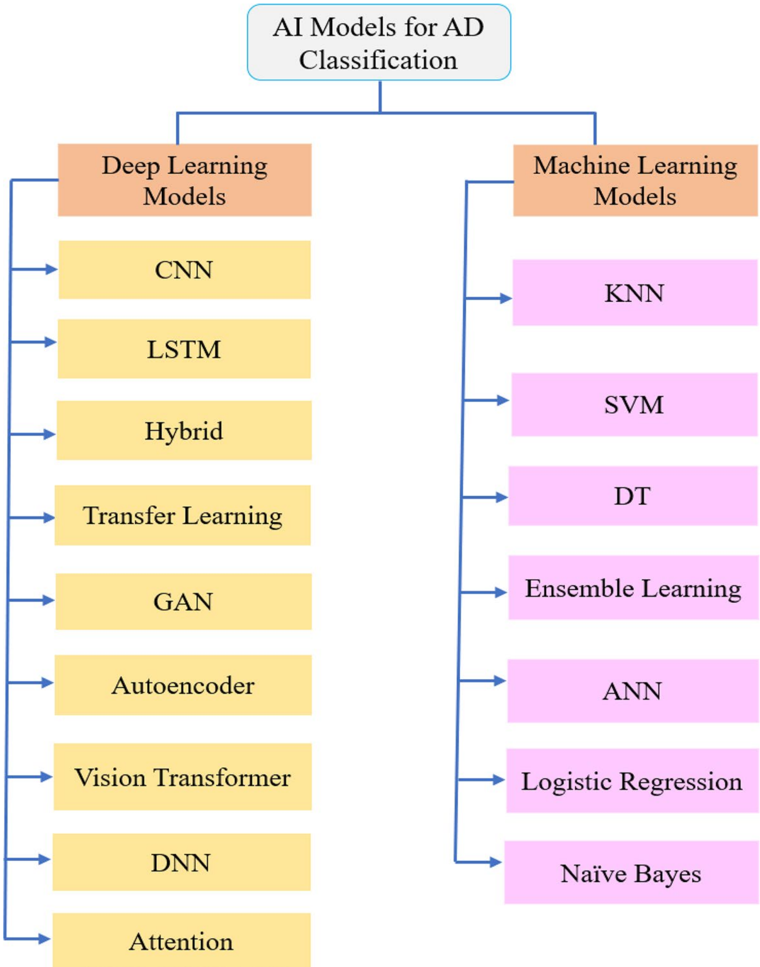

Fig. 4 Most common AI models for AD classification

1. Early Diagnosis and Risk Assessment: DL models enable early detection of MCI, a precursor to AD, from structural and functional brain imaging. The impact of this application is used to provide timely interventions, slowing disease progression. Also, it aids in personalized treatment planning (Shuvo et al. 2022).
2. Clinical Decision Support Systems (CDSS): Automated diagnostic tools powered by DL assist clinicians in identifying AD from MRI, PET, or CT scans with high accuracy. The effect of this application is utilized to reduces diagnostic errors and inter-observer variability. Additionally, it saves time for clinicians by automating feature extraction and classification (Duarte de Almeida and Oliveira 2019).
3. Neuroimaging Biomarker Identification: DL models help identify biomarkers from imaging data, such as hippocampal atrophy that correlate with AD. The benefit of this application is to enhance understanding of disease pathology and support the development of targeted therapies (Skolariki et al. 2020).
4. Disease Progression Prediction: Predicting the transition from MCI to AD using sequential imaging data analyzed by DL models (e.g., hybrid CNN-RNN architectures). The impact of this application is utilized to help in monitoring disease progression and inform long-term care strategies and research studies (Guarín et al. 2024).
5. Personalized Treatment Monitoring: DL analyzes neuroimaging and clinical data to assess the performance of treatments and interventions. The effect of this application is used to enable dynamic adjustments to treatment plans and facilitate real-time monitoring of therapeutic outcomes (Koutkias et al. 2010).
6. Cognitive Function Prediction: Predicting cognitive scores (e.g., MMSE) based on imaging and non-imaging data using DL regression models. The benefit of this application is to offer a non-invasive proxy for cognitive testing and support routine clinical evaluations (Zhang et al. 2013).
7. Drug Development and Clinical Trials: DL aids in patient stratification by accurately classifying participants into healthy, MCI, and AD groups, ensuring homogeneous study cohorts. It effects on reduceing heterogeneity in clinical trials and facilitating drug discovery and efficacy testing (Sonka and Grunkin 2002).
8. Public Health Screening: Large-scale, automated AD screening using DL-powered platforms integrated with imaging tools (e.g., mobile MRI units or cloud-based systems). This application is used to expand access to diagnostic services in underserved regions and support epidemiological studies and public health initiatives (Fletcher et al. 2017).
9. Integration with Wearable Devices: Combining DL models with data from wearable sensors that track sleep, physical activity, or speech patterns to predict cognitive decline. The impact of this application is used to offer non-invasive, continuous monitoring of at-risk individuals. Also, it promotes early-stage intervention (Alattar and Mohsen 2023).
10. Real-Time Diagnosis in Telemedicine: DL models integrated into telemedicine platforms analyze remotely acquired imaging and clinical data to provide diagnostic insights. The effect of this application is utilized to increase diagnostic reach, especially in rural areas and reduce patient burden of traveling for clinical assessments (Yan and Song 2010).
11. Cross-Modality Analysis: DL combines data from various imaging modalities (e.g., MRI and PET) for more comprehensive diagnostics. There are two effects for this

application. The first one is improving classification accuracy, while the second effect is providing a holistic view of structural and functional changes in the brain (Feng et al. 2021).

12. Educational Tools for Medical Training: DL-powered visualizations (e.g., Grad-CAM, saliency maps) are used as teaching tools for medical students and radiologists to interpret AD-related changes. There are twp benefits for this application: enhancing learning through real-world data and reducing the gap between practical application and theoretical knowledge (Sakuma 2013).
13. Synthetic Data Generation: Generative Adversarial Networks (GANs) create synthetic neuroimaging data to augment limited datasets. It addresses data scarcity and class imbalance issues and facilitates training of more robust DL models (Failed 2023). Figure 5 shows several applications of Alzheimer disease with DL.

4 Datasets and preprocessing techniques for Alzheimer disease detection

There are publicly available datasets in the field of AD, such as ADNI (Alzheimer's Disease Neuroimaging Initiative) that are a comprehensive dataset with MRI and PET images, OASIS (Open Access Series of Imaging Studies) that focus on aging and AD detection, and AIBL (Australian Imaging, Biomarkers, and Lifestyle) dataset that provides longitudinal neuroimaging data.

The success of DL models in Alzheimer's disease classification heavily relies on high-quality, well-labeled datasets. Several public datasets have become the benchmark for researchers in this field, allowing for reproducible results and fostering collaboration among researchers.

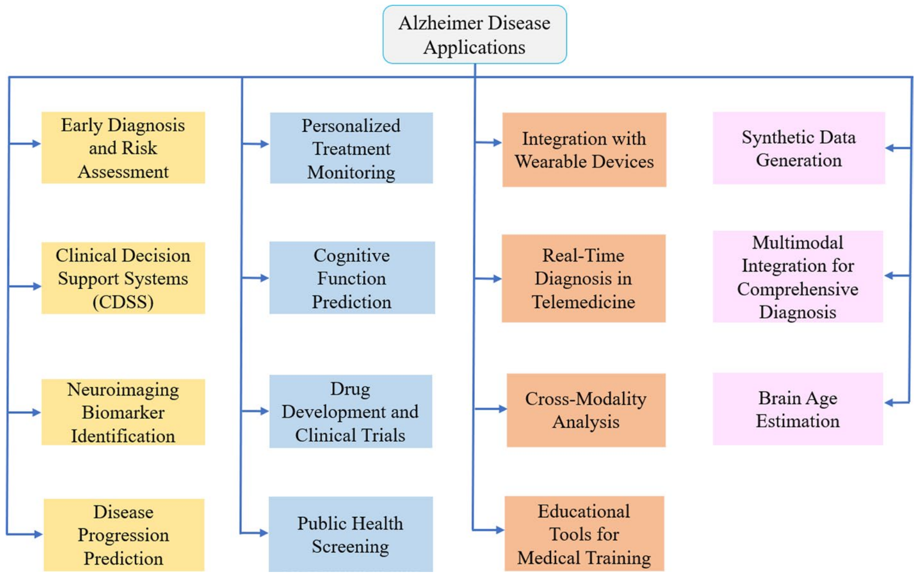

Fig. 5 Applications of Alzheimer disease with deep learning

4.1 Datasets for Alzheimer disease detection

The following datasets are widely used in Alzheimer's disease detection:

1. Alzheimer's Disease Neuroimaging Initiative (ADNI) dataset (Chu and Gebre-Amlak 2021) is a widely used resource for AD classification, containing MRI, PET scans, and clinical data from subjects at various disease stages. It provides both cross-sectional and longitudinal data, aiding in classification and disease progression tracking. Deep learning models trained on ADNI have shown high accuracy, and its inclusion of genetic data supports multimodal classification approaches.
2. Open Access Series of Imaging Studies (OASIS) dataset (Marcus et al. 2010) provides MRI scans and clinical data for individuals with AD, MCI, and healthy controls. It is valuable for studying early Alzheimer's detection, as it includes MCI patients. Research using OASIS has shown deep learning models can identify early signs of AD, such as brain atrophy and cortical thinning.
3. Australian Imaging, Biomarkers, and Lifestyle Study (AIBL) dataset (Bloch and Friedrich 2019) provides a valuable resource for AD research, offering MRI scans, PET data, and clinical information for individuals at different stages of cognitive decline. It is particularly beneficial for studying the correlation between lifestyle factors, biomarkers, and brain changes in Alzheimer's disease. Research leveraging this dataset has shown how deep learning models can integrate lifestyle data and imaging data to predict the onset of Alzheimer's, highlighting the potential for preventive strategies.
4. Indian Cognitive and Health Data Repository (ICHDR) dataset (PJKRKSN and SDP 2023) For regional studies, the ICHDR dataset provides a rich source of data for Alzheimer's disease classification in the Indian population. This dataset includes brain MRI images, neuropsychological assessments, and genetic information. Models trained on ICHDR can be used to develop region-specific detection tools, as factors like cultural practices and genetics may influence the presentation of AD.
5. Harvard Medical School (HMS) dataset (Assam et al. 2021) has features T2-weighted brain MRI images with dimensions of 265×256 pixels. This collection comprises 613 images, categorized into two classes related to Alzheimer's disease: 27 images fall under the normal class, representing two cases, while 513 images belong to the abnormal class, covering forty cases. The abnormal class encompasses conditions associated with degenerative, cerebrovascular, and provocative diseases. A unique aspect of the HMS dataset is its focus on individuals who are non-cognitively impaired at the outset, making it especially significant for studies targeting the early detection of Alzheimer's. The dataset spans multiple years, offering a longitudinal perspective that enables researchers to observe changes in brain structure and function over time, capturing the disease's early progression.

HMS promotes collaboration and inclusivity by offering its dataset freely to the research community. However, as a newer initiative compared to established datasets like ADNI and OASIS, HMS may currently have a more limited data volume, potentially restricting the depth of analysis. Furthermore, while the dataset is publicly accessible, specific details regarding data types and access protocols might not be as well-documented or user-friendly as those provided by ADNI and OASIS.

Table 1 Comparisons of different datasets

References	Dataset	Diversity	Size (AD patients)	Modality	Strengths	Limitations
Chu and Gebre-Amlak (2021)	ADNI	Mostly White/Caucasian	~600–800	MRI, PET, fMRI	Comprehensive, multi-modal, large-scale	Longitudinal data increases complexity
Marcus et al. (2010)	OASIS	Mostly White	~200	MRI	Open access, focused on aging-related studies	Smaller dataset compared to ADNI
Bloch and Friedrich (2019)	AIBL	Mostly White (Australian population)	~250–300	MRI, PET	Includes cognitive scores, longitudinal data	Geographically limited population
P. J. K, R. K, S. N and S. D. P (2023)	ICHDR	More diverse, includes Asian populations	~400–500	MRI	Includes brain neuropsychological assessments and genetic information	Small dataset
Assam et al. (2021)	HMS	Ethnically diverse (African American, Hispanic, White)	~1000	MRI	High resolution, large scale	Only consists of two classes

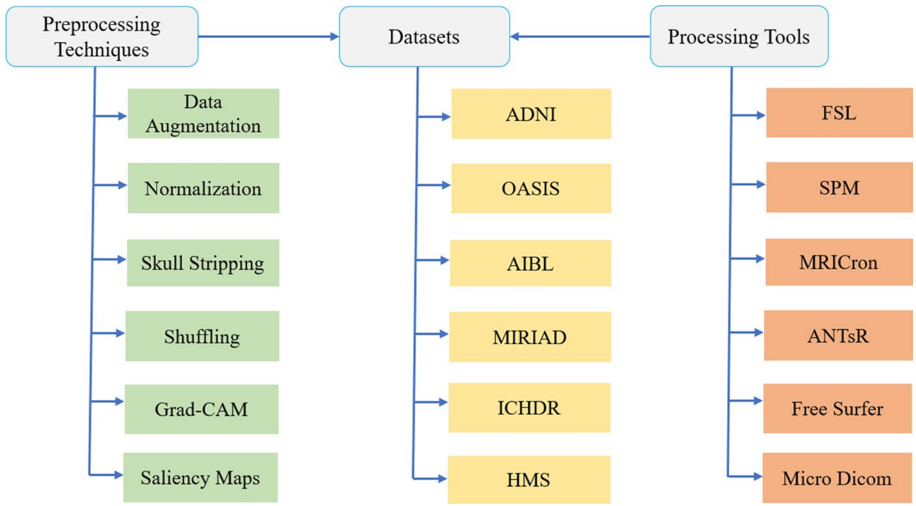

Fig. 6 Different datasets, processing techniques, and tools used with AD

Table 1 illustrates some different dataset comparisons. Figure 6 illustrates different datasets, preprocessing techniques, and processing tools used with AD.

All these mentioned datasets need different preprocessing techniques such as normalization and scaling to ensure consistent intensity across scans, data augmentation to expand dataset size, improving model robustness, and shuffling to prevent data leakage and enhances generalization. So, preprocceing techniques are very important to obtain a high effectiveness of AI models.

4.2 Preprocessing techniques for Alzheimer disease detection

The following preprocessing techniques are utilized in Alzheimer’s disease classification:

1. **Data augmentation** is a technique used to artificially enlarge the size and diversity of datasets by employing different transformations to the existing data. This process is especially valuable in scenarios where the available data is limited. For images, common augmentation methods include flipping, rotating, zooming, cropping, adding noise, and color shifting. For text, techniques such as random insertion, synonym replacement, deletion, or translation are used. The primary goal of data augmentation is to enhance a model's prediction ability, reduce overfitting, and improve robustness by exposing it to a broader range of variations in the input data (Nanthini et al. 2023).
2. **Normalization** is a preprocessing step that changes the scale of data features to bring them within a consistent range, ensuring comparability. In image processing, normalization typically involves scaling values of pixels from 0–255 to a range such as 0–1 or -1 to 1. This technique is beneficial because it speeds up the training process, ensures a more uniform feature distribution, and helps machine learning models converge faster. A common normalization formula involves subtracting the average and dividing by the standard deviation of the dataset (Huang et al. 2023).
3. **Skull stripping** is a preprocessing step in medical imaging, particularly for MRI scans. This technique involves removing the skull and other non-brain tissues from the image, leaving only the brain region. Skull stripping enhances the focus on brain-specific areas by eliminating irrelevant features and noise, which, in turn, enhances the efficiency of models analyzing these images. Methods for skull stripping include threshold-based techniques, region-growing algorithms, and advanced deep learning-based automated approaches (Singh et al. 2023).
4. **Shuffling** is the process of randomizing the order of data samples before feeding them into a machine learning model. This step is crucial in preventing the model from learning biases or patterns that are specific to the order of the data. By shuffling, we ensure that the training process does not inadvertently favor certain sequences or classes, enhancing the model's ability to predict. Shuffling is often used in conjunction with stochastic gradient descent (SGD) during model training to maintain randomness and variability in the input (Nehal et al. 2023).
5. **Gradient-weighted class activation mapping** (Grad-CAM) is a visualization technique designed to provide insights into the decision-making process of deep learning models, particularly CNNs. It works by leveraging the gradients flowing into the last convolutional layer of the model to compute a weighted activation map. This map is then used to create a heatmap that highlights the regions in an input image that contribute most to the model’s prediction. Grad-CAM is especially useful for debugging, understanding model behavior, and improving interpretability by revealing which parts of an image influenced the decision (Quach et al. 2023).
6. **Salience maps** are another visualization tool that highlights the most important regions of an input image influencing a model's output. These maps are generated by computing

the gradients of the model's output with respect to the input image, showing the sensitivity of the output to changes in specific input features. Salience maps are widely used in tasks like feature importance analysis, object detection, and attention mechanisms, providing a clearer understanding of the model's focus and decision-making process (Mukherjee et al. 2015). Table 2 shows comparisons of several preprocessing techniques.

4.3 Explanation of model interpretability techniques

Techniques like SHAP and LIME provide powerful ways to understand and explain AI model decisions, increasing clinician trust and promoting AI adoption in healthcare. By integrating these methods into decision-support systems, we can improve diagnostic accuracy and treatment outcomes while ensuring greater acceptance by both doctors and patients.

4.3.1 Shapley additive explanations (SHAP)

SHAP is based on Shapley values from game theory, which analyze the contribution of each feature to a model's prediction. The core idea is to compute the average impact of each feature across all possible feature orderings, providing both global and local interpretability of the model. The SHAP has three advantages: the first one, it provides a consistent and fair explanation of feature importance. Also, it can be applied to deep learning models, random forests, and regression models. Further, it offers global insights into the model's behavior as well as local explanations for individual predictions.

Table 2 Comparisons of several preprocessing techniques

References	Technique	Benefits	Challenges
Nanthini et al. (2023)	Data Augmentation	Expands dataset size, improves generalization	Risk of introducing unrealistic transformations
Huang et al. (2023)	Normalization	Reduces inter-subject variability in intensity	May lead to loss of contrast in critical regions
Singh et al. (2023)	Skull Stripping	Removes non-brain tissues for cleaner inputs	Time-consuming, requires robust automation tools.
Nehal et al. (2023)	Shuffling	Prevents data leakage, ensures balanced training	Adds computational cost during each epoch
Quach et al. (2023)	Grad-CAM	Highlights critical regions	Can mislocalize subtle features
Mukherjee et al. (2015)	Saliency Maps	Visualizes feature contributions	Overcoming noise in input data

4.3.2 Local interpretable model-agnostic explanations (LIME)

LIME focuses on local interpretability by creating a simplified model (e.g., linear regression) to approximate the predictions of the original model in a small neighborhood around a specific data point. This is achieved by generating perturbed data and analyzing how feature variations affect predictions. The advantages of the Lime are: Fast and effective in explaining individual predictions, model-agnostic, meaning it can be applied to any machine learning model, and helps in understanding why a model made a particular decision for a specific patient.

4.3.3 Role of model interpretability in enhancing clinician trust and adoption

In healthcare, clinicians rely on logical and explainable reasoning when making critical decisions. If they understand how and why a model provides a certain diagnosis, they will have more confidence in using it. The ability to justify predictions is crucial, as medical professionals need to ensure that AI-driven recommendations align with established clinical knowledge and patient history. Techniques like SHAP and LIME enhance trust by offering clear explanations of model decisions, allowing doctors to compare AI-generated insights with their own expertise. This transparency reassures clinicians that AI is a supportive tool rather than an unpredictable "black-box" system. One of the biggest barriers to AI adoption in healthcare is the concern that models operate as opaque "black boxes," making decisions without clear reasoning. Interpretable AI models address this issue by revealing which factors contribute to specific predictions, increasing transparency and credibility.

5 Recent studies in AD detection

There are recent studies in the field of AD presented (Slimi et al. 2024; Ching et al. 2024; Alsubaie et al. 2024; Jo et al. 2019; Lu et al. 2019; Iriondo et al. 2203; Raza et al. 2023; Gnanasegar et al. 2020). In (Slimi et al. 2024), the hybrid model integrating Xception and DenseNet121 architectures has demonstrated outstanding performance, achieving an accuracy of 99.85% for AD classification. This success is attributed to the combination of Xception's efficient depthwise separable convolutions and DenseNet121's connected layers, which enhance feature extraction and improve the model's representational capacity. The used dataset is preprocessed MRIs. The use of the Synthetic Minority Over-sampling Technique (SMOTE) effectively addressed class imbalance, ensuring better generalization across underrepresented categories. However, this hybrid model comes with notable challenges. The integration of two complex architectures results in high computational cost and increased processing overhead, which poses difficulties for real-world scalability. Additionally, the model's complexity may limit its adaptability to settings with constrained computational resources, highlighting the trade-off between performance and practicality.

In (Ching et al. 2024), the EfficientNet-B0 model, leveraging transfer learning, offers a lightweight and efficient solution for AD detection, reaching an accuracy of 87.17%. Its streamlined architecture makes it particularly appropriate for deployment on systems with limited computational resources, like edge devices or smaller healthcare facilities. Despite its advantages, the model has limitations in handling complex feature representations due to its reduced network depth, which can impact its ability to discern subtle patterns in medical imaging data. Additionally, its performance is relatively lower compared to deeper architectures, especially in multi-class classification tasks where a greater level of feature discrimination is required. These trade-offs highlight the balance between model efficiency and accuracy, with EfficientNet-B0 excelling in resource-constrained environments but facing challenges in more demanding classification scenarios. The used datasets are MRI scans divided into four classes (non-demented to severe AD).

In (Alsubaie et al. 2024), Capsule Networks (CapsNets) have shown promising potential in Alzheimer's disease classification because of their robust feature representation capabilities. CapsNets excel in capturing spatial hierarchies and relationships between features, which is particularly beneficial for analyzing complex brain imaging data. Their architecture also helps reduce overfitting, even when applied to datasets with variations in segmentation, enhancing model robustness. Despite these advantages, CapsNets face significant challenges, including high computational demand and slower training times compared to traditional CNNs. Additionally, their scalability remains limited, as handling large-scale datasets efficiently is difficult. These trade-offs make CapsNets a compelling, yet computationally intensive, alternative for medical imaging tasks.

In (Jo et al. 2019), deep boltzmann machines (DBMs) have been applied to AD detection by integrating multi-modal neuroimaging data, demonstrating their effectiveness in predicting the conversion from MCI to AD. By leveraging data from multiple sources, such as MRI, PET, and clinical measures, DBMs capture complementary features that enhance prediction accuracy and provide a holistic view of patient conditions. However, their effectiveness depends heavily on the availability of high-quality, multi-modal datasets, which are often challenging to collect in clinical practice. Additionally, the training of DBMs is computationally intensive, requiring significant processing power and time. These limitations, while notable, are balanced by their ability to process and synthesize complex, diverse datasets, offering valuable insights in early diagnosis and disease progression modeling.

In (Lu et al. 2019), the CNN-LSTM hybrid model combines the spatial feature extraction capabilities of CNNs with the temporal sequence processing strength LSTM networks, achieving a high accuracy of 98.5% on segmented datasets for Alzheimer's disease detection. This integration allows the model to capture both spatial patterns from medical imaging and temporal dependencies, such as progressive changes in brain scans over time. However, this approach necessitates advanced pre-processing techniques, including segmentation and normalization, to optimize data quality and ensure effective model training. Additionally, the hybrid architecture introduces increased training complexity and higher memory requirements, making it more computationally demanding compared to simpler models. Despite these challenges, the CNN-LSTM hybrid model is a powerful tool for analyzing time-sequenced medical data in Alzheimer's research. This work utilized two MRI datasets with Alzheimer's disease stages.

In (Iriondo et al. 2203), the DeepAD model leverages advanced DL techniques for predicting AD progression by integrating clinical information and 3D MRI scans from multiple cohorts. A key strength of DeepAD lies in its ability to address domain adaptation and inter-study biases through adversarial training, which enhances its robustness across diverse datasets. The model effectively combines clinical data with imaging features, utilizing mutual information loss to promote domain generalization and improve prediction consistency across varying data sources. Despite its strengths, DeepAD's multi-network architecture makes it computationally intensive, demanding significant processing power and memory resources. Furthermore, its reliance on high-quality multi-modal datasets can limit its applicability in settings where such data are unavailable, posing challenges for universal deployment. Nonetheless, DeepAD represents a significant step forward in integrating diverse data types for accurate disease progression prediction.

In (Raza et al. 2023), the DenseNet and ResNet-based classification approach combines the strengths of DenseNet-169 and ResNet-50 architectures for robust feature learning, utilizing datasets such as OASIS and other publicly available MRI repositories. This dual-model strategy has proven highly effective in detecting Alzheimer's disease, particularly in its early stages, achieving high accuracy levels. DenseNet's efficient feature reuse and ResNet's residual learning help capture intricate patterns in neuroimaging data. However, a notable limitation of these models is their susceptibility to performance degradation when dealing with imbalanced datasets. Without the application of preprocessing techniques such as oversampling or data augmentation, the models may struggle to generalize effectively, especially when minority classes are underrepresented. These challenges underscore the significance of preprocessing in improving model efficiency for real-world applications.

In (Gnanasegar et al. 2020), an integration of a LSTM model with a Boruta based feature selection presented for a classification task. This model trained on the OASIS MRI dataset for Alzheimer's disease. The Boruta algorithm, a wrapper-based feature selection method, was employed to identify Clinical Dementia Rating (CDR) and Mini Mental Status Examination (MMSE) scores as key predictors of dementia. The LSTM-based model classified subjects as demented or non-demented, achieving an impressive 94% accuracy. A key advantage of our approach is its ability to mitigate overfitting, ensuring robust and reliable performance. Table 3 summarizes several recent studies on different DL models for AD classification.

6 Comparison of AI approaches versus traditional AD diagnostic methods

Traditional methods (MMSE, CDR, PET) remain valuable, but have limitations in early detection, cost, and scalability, while AI models outperform them in accuracy, efficiency, and early detection, especially when using deep learning on MRI and PET data. AI could augment traditional methods rather than replace them, assisting clinicians in making faster, more precise diagnoses.

Traditional AD diagnostic methods such as MMSE, CDR, and PET. The MMSE is a simple 30-question cognitive test used to assess memory, attention, and problem-solving. However, it lacks sensitivity to early-stage AD and is influenced by education level and language skills, while the CDR is a clinician-led structured interview evaluating six domains (memory, orientation, problem-solving, etc.). Results depend on expert interpretation, making it prone to variability. The PET scans are highly accurate, but expensive and not widely available. PET detects amyloid-beta and tau protein deposits, biomarkers of AD.

AI-based models can analyze MRI/PET changes and detect neurodegeneration years before cognitive symptoms appear. Some models achieve > 90% accuracy, compared to MMSE's ~ 80%. DL models use MRI scans to objectively quantify brain atrophy in key regions (e.g., hippocampus), reducing subjectivity. AI models are more accessible and less

Table 3 Several Recent Studies on different DL models for Alzheimer's disease classification

Study	Model	Accuracy (%)	Strengths	Weaknesses
Slimi et al. (2024)	Xception + DenseNet121	99.85	Handles imbalance, excellent accuracy	High computational cost
Ching et al. (2024)	EfficientNet-B0	87.17	Lightweight, efficient	Limited feature depth
Alsubaie et al. (2024)	Capsule Networks (CapsNet)	93.50	Robust feature representation	High computational demand
Jo et al. (2019)	DBM with Multimodal Data	98.8	Effective multi-modal integration	Requires high-quality datasets
Lu et al. (2019)	CNN-LSTM Hybrid Model	98.50	Captures spatial–temporal features	Increased preprocessing and complexity
Iriondo et al. (2203)	DenseNet + Adversarial Networks	–	Robust to domain shifts, multi-modal learning	High computational cost, data-dependent
Raza et al. (2023)	DenseNet169	97.84	Feature richness	Class imbalance issues
Gnanasegar et al. (2020)	LSTM	94	mitigate overfitting and Boruta feature selection	Low accuracy, limited analysis

costly, while still achieving high diagnostic performance. Some DL models trained on MRI data achieve accuracy comparable to PET-based assessments. Table 4 shows a comparison of AI Approaches versus Traditional AD Diagnostic Methods.

7 Real-world implementation of AI models and dataset adaptability

In this Section, we present some real-world implementations of AI models in clinical settings and their adaptability to different datasets.

7.1 AI in medical imaging

Application: AI models like Google's DeepMind and IBM Watson Health analyze X-rays, MRIs, and CT scans to detect diseases such as cancer, pneumonia, and brain hemorrhages. Model: Deep learning-based Convolutional Neural Networks (CNNs), use case: radiology and pathology. Adaptability: AI models trained on large, diverse datasets like the ChestX-ray14 dataset (NIH) can be fine-tuned with local hospital data to improve specificity. Also, transfer learning enables adaptation to new datasets with fewer labeled images, allowing hospitals in different regions to deploy these systems efficiently.

Table 4 Comparison of AI approaches versus traditional AD diagnostic methods

Aspect	Traditional AD diagnostic methods	AI-based approaches
Type of data used	Clinical assessment (MMSE, CDR), neuroimaging (PET, MRI), genetic tests	Multimodal data integration (MRI, PET, genetic, behavioral, electronic health records)
Subjectivity	High (depends on clinicians expertise)	Low (objective, data-driven)
Early detection ability	Limited, often detects moderate to severe cases	Superior, can detect preclinical AD before symptoms appear
Time and Cost	Time-consuming, costly (especially PET scans)	Faster Processing, reduces costs over time
Scalability	Limited by clinicians availability and resources	Highly scalable, can analyze large datasets automatically
Interpretability	Well understood by clinicians	Improving with explainable AI (XAI), but still a challenge
Longitudinal tracking	Requires repeated clinical visits	AI methods can track cognitive decline overtime using DL
Accuracy	Moderate Accuracy (MMSE ~ 8%, CDR: clinical dependent)	Higher accuracy (over 90% for some AI models)

7.2 AI for predictive analytics in ICU

Application: AI-powered sepsis watch system at Duke University analyzes vital signs, lab results, and medical history to predict sepsis risk in ICU patients. Model: Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks, use case: early sepsis detection. Adaptability: the model integrates data from Electronic Health Records (EHR) such as MIMIC-III (MIT). Also, continuous training on new patient data ensures adaptability to different populations and hospital workflows.

7.3 AI in personalized treatment planning

Application: IBM Watson for Oncology assists doctors by recommending personalized treatment plans based on patient history, clinical trials, and medical literature. Model: Reinforcement Learning (RL) and Decision Support Systems, use case: oncology (cancer treatment). Adaptability: It uses Natural Language Processing (NLP) to analyze global and local datasets. Also, customization with regional treatment guidelines ensures adaptability to different healthcare environments.

7.4 AI-powered robotic surgery

Application: The da Vinci Surgical System enhances precision in robotic-assisted surgeries by analyzing real-time data. Model: Computer Vision & Reinforcement Learning, use case: minimally invasive surgeries. Adaptability: AI learns from past surgical procedures and adapts to different patient anatomies. Also, federated learning allows hospitals to share anonymized data to improve model performance without compromising patient privacy.

When evaluating the potential clinical adoption of different models, several key factors must be considered alongside accuracy: clinical adoption of models depends on interpretability, scalability, and efficiency. Explainable AI (XAI) methods enhance trust by providing clear insights into model decisions. Scalability factors, such as computational cost and EHR integration, affect real-world deployment, with lighter or hybrid models being more feasible. Balancing accuracy with efficiency is crucial, as high-performance models with excessive computational demands may be impractical for widespread clinical use.

7.5 Generalizability of AI models in healthcare

Generalizability refers to an AI model's ability to maintain high performance when applied to new datasets, different hospitals, and diverse patient populations. Challenges to generalizability as follows:

Data Distribution Shift: AI models trained on one dataset may not perform well on data from another region due to variations in imaging protocols, demographics, and clinical practices. Additionally, limited Representation: many AI models are trained on datasets from a specific region, leading to biases when applied globally. Also, overfitting to training data: deep learning models might memorize patterns rather than learning generalizable features.

7.6 Scalability of AI models in clinical workflows

Scalability refers to an AI model's ability to handle increasing amounts of data, users, and healthcare facilities while maintaining efficiency. Challenges to scalability as follows: computational Costs: Large AI models require significant processing power, limiting deployment in resource-constrained hospitals. Additionally, integration with EHR Systems: Hospitals use different Electronic Health Record (EHR) formats (e.g., Epic, Cerner, Meditech), making AI integration complex. Also, regulatory and compliance Hurdles: AI models must meet local regulations (e.g., HIPAA, GDPR) before large-scale deployment.

7.7 Deployment challenges of AI models

Despite the potential of AI to revolutionize healthcare, deploying AI models in real-world clinical environments presents several challenges. These challenges range from technical and regulatory issues to ethical and operational hurdles.

1. Data-Related Challenges: healthcare data is sensitive and protected by regulations like HIPAA (USA), GDPR (Europe), and other regional laws. AI models require large datasets, making compliance difficult. Also, healthcare data is stored in various formats across hospitals (EHR systems like Epic, Cerner, and Meditech) with different structures. Lack of standardization makes integration difficult.
2. Model-Related Challenges: many AI models, especially deep learning models, function as "black boxes," making it difficult for clinicians to understand how decisions are made. Also, AI models may become less accurate over time due to changes in medical practices, equipment, or patient demographics (a phenomenon known as model drift).

3. Operational & Infrastructure Challenges: AI models often require significant modifications to existing clinical workflows, leading to resistance from healthcare professionals. Additionally, many hospitals lack in-house AI expertise, making it difficult to maintain and update AI models.
4. Regulatory and Ethical Challenges: AI models must comply with stringent regulations before they can be used in clinical practice (e.g., FDA approval in the USA, CE marking in Europe). Also, patients may be reluctant to trust AI-based diagnoses or treatment recommendations.

8 Overfitting problem in DL and mitigation methods

8.1 Overfiting description

Overfitting occurs when a trained model becomes overly adapted to the training data, leading to excellent performance on training data but poor generalization to test or unseen data. This means the model memorizes patterns specific to the training set rather than learning general trends that can be applied to new data.

Overfitting typically happens when the model is too complex, containing too many parameters relative to the dataset size. It also occurs when the model fits the training data too tightly, capturing noise or irrelevant patterns instead of useful trends. Additionally, if the model relies on unnecessary or irrelevant features, it becomes ineffective when tested on new datasets.

There are several signs that indicate a model is overfitting. The most common sign is an extremely high accuracy on training data but a significant drop in performance on test data. Another indicator is a large gap between training accuracy and test accuracy, showing that the model is not generalizing well. If a model is excessively complex without improving real-world predictions, it is also likely to be overfitting.

8.2 Mitigation methods against overfiting

To reduce overfitting, one effective method is increasing the dataset size. More data helps the model generalize better instead of memorizing training samples. When additional data is not available, data augmentation techniques, such as image rotation, flipping, brightness adjustments, and adding noise, can artificially expand the dataset. Another approach is to simplify the model. Reducing the number of layers in neural networks or selecting a less complex algorithm can prevent overfitting. Similarly, reducing the number of parameters in the neural network architecture helps in making the model more generalizable. Cross-validation is also a useful technique for mitigating overfitting. K-Fold Cross-Validation ensures that the model is tested on multiple subsets of data, reducing the likelihood of overfitting by providing a better estimate of its performance on unseen data. Regularization techniques play a crucial role in preventing overfitting. L1 regularization (Lasso) penalizes the absolute values of parameters, forcing some of them to become zero and effectively selecting only important features. L2 regularization (Ridge), on the other hand, penalizes the squared values of parameters, reducing their magnitude without eliminating them, which results in a more stable model. For neural networks, dropout is an effective regularization method that randomly disables neurons during training to prevent the model from depending too much on specific features. Feature selection is another key strategy. Removing unnecessary or low-impact features reduces the complexity of the model, making it less prone to overfitting. Dimensionality reduction techniques such as Principal Component Analysis (PCA) can help retain only the most significant features while discarding irrelevant ones. Adding noise to training data is another way to prevent overfitting. By introducing random variations, the model learns to generalize better instead of relying on minor patterns that may not be present in real-world data. This is especially useful in neural networks to enhance their robustness. Lastly, early stopping is a widely used technique to avoid overfitting. By monitoring the model's performance on a validation set, training can be stopped when validation accuracy starts to decline, preventing the model from memorizing the training data instead of generalizing.

9 Medical modalities used with AD detection

In this Section, we explain different modalities used for Alzheimer disease detection. Each modality has unique strengths, making them complementary tools for diagnosis and research in various medical fields, especially neurology and oncology.

Computed tomography (CT) is a medical imaging technique that utilizes X-rays to produce detailed cross-sectional images of the body. The patient lies in a rotating scanner where X-rays capture data, which a computer reconstructs into 2D or 3D images. CT scans are commonly used to detect tumors, fractures, and internal bleeding, making them invaluable in emergency diagnostics due to their speed. They give high-resolution images of both bones and soft tissues, but CT scans involve exposure to ionizing radiation and are less effective than MRI for visualizing soft tissue contrast (Gao and Hui 2016).

Structural magnetic resonance imaging (sMRI) is a non-invasive imaging technique that utilizes roubst magnetic fields to produce highly detailed images of body structures. It is particularly useful for visualizing brain anatomy, helping detect abnormalities such as tumors or atrophy. sMRI plays a key role in studying structural changes in neurodegenerative diseases like Alzheimer's. It offers excellent soft tissue contrast without exposing patients to ionizing radiation. However, it is time-consuming, expensive, and unsuitable for individuals with metal implants or severe claustrophobia (SSCS and B U 2023).

Positron emission tomography (PET) is a functional imaging technique that reveals metabolic or biochemical activity in tissues. A radioactive tracer is injected into the body, and as it decays, the PET scanner detects the emitted gamma rays to generate images. PET scans are widely used to identify cancer, monitor its spread, and assess brain activity in conditions like Alzheimer's. They are also effective in evaluating heart function and blood flow. While PET provides valuable functional data and is sensitive to abnormal cellular activity, it involves exposure to radioactive materials and has lower spatial resolution compared to MRI or CT (Salah et al. 2024).

Diffusion tensor imaging (DTI) is a form of MRI that maps the diffusion of water molecules within tissues, primarily to study neural pathways in the brain. By measuring the direction of water diffusion, DTI visualizes white matter tracts and brain connectivity. It is particularly useful for diagnosing conditions like traumatic brain injury, multiple sclerosis, and stroke, as well as for research in neural development and disorders. DTI is non-invasive and effective for assessing white matter integrity but is sensitive to motion artifacts and requires complex data analysis (Sang and Li 2024).

Functional magnetic resonance imaging (fMRI) measures brain activity via detecting changes in blood flow associated with neural activity (Yang et al. 2024). This technique uses the Blood Oxygen Level-Dependent (BOLD) signal to pinpoint active brain regions during specific tasks or rest. fMRI is a vital tool in cognitive neuroscience, mapping functions like motor and sensory areas, and aiding pre-surgical planning for brain tumor removal. While it is non-invasive and offers high spatial resolution, fMRI is an indirect measure of neuronal activity, relying on blood flow changes, and is highly sensitive to motion artifacts.

Figure 7 illustrates the percentage of utilization of each different imaging modality with Alzheimer Disease.

10 Evaluation metrics used in AD detection

In Alzheimer's disease (AD) classification, assessing the performance of deep learning models is essential to ensure their reliability and clinical applicability. Various evaluation metrics are employed to assess model performance, each providing unique insights into how well a model distinguishes between healthy and AD patients. There are many standard metrics for model assessing include Precision, Recall, Accuracy, F1-Score to quantify classification performance, Receiver Operating Characteristic-Area Under Curve (ROC-AUC) to analyze discriminative ability, and also various loss Metrics to evaluate convergence during training. Each metric has its advantages and is suitable for different types of analysis, based on the nature of the dataset, the issue at hand, and the consequences of model errors. The following are some of the most commonly used metrics, along with the following Equations (Mohsen et al. 2023b; Chicco et al. 2021; Patil and Nisha 2021).

10.1 Accuracy

It is one of the most basic and oftenly utilized metrics in classification tasks. It represents the proportion of correctly predicted samples (both positive and negative) to the overall number of instances. In the context of Alzheimer's classification, accuracy calculates how often the model correctly classifies both the Alzheimer's patients and healthy individuals. Equation (1) represents the accuracy.

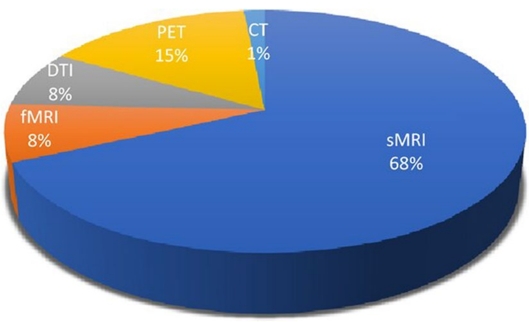

Fig. 7 Percentage of utilization of each different imaging modality with Alzheimer disease

$$Accuracy = \frac{TP + TN}{TP + FP + TN + FN} \tag{1}$$

where *TP*=True Positives (correctly predicted AD cases), TN=True Negatives (correctly predicted healthy cases), FP=False Positives (healthy cases predicted as AD), and FN=False Negatives (AD cases predicted as healthy). While accuracy is a valuable measurement, it can be misleading when dealing with unbalanced datasets, where the number of healthy individuals far outweighs the AD patients.

10.2 Precision

In medical classification tasks like AD detection, where the consequences of FPs and FNs are significant, more measurements like precision, recall, and F1-score are preferred. Precision (also known as PPV) measures the proportion of true positive predictions between all the positive predictions made by a model. This is especially important when false positives (misdiagnosing healthy individuals as AD patients) are a concern. It is represented by Eq. (2).

$$Precision = \frac{TP}{TP + FP} \tag{2}$$

10.3 Recall

It is also known as sensitivity, indicates the proportion of actual positive cases that were correctly identified by a model. Recall is particularly important when false negatives (failing to identify AD patients) are a concern. Recall is calculated by Eq. (3).

$$Recall = \frac{TP}{TP + FN} \tag{3}$$

10.4 F1-score

It is a harmonic mean of precision and recall, providing a balance between them. The F1-score is especially beneficable when the dataset is unbalanced, as it provides a more balanced view of the model's efficiency than accuracy alone. Studies in AD detection often prioritize F1-score because misdiagnosing AD patients (false negatives) can be more dangerous than misclassifying healthy individuals. It is determined through Eq. (4).

$$F1 - score = 2 \times \frac{Precision \times Recall}{Precision + Recall} \tag{4}$$

10.5 Area under the receiver operating characteristic curve (AUC-ROC)

It is another popular metric, particularly in binary classification tasks like Alzheimer's disease detection. It plots the True Positive Rate (Recall) against the False Positive Rate (FPR), which is defined in Eq. (5).

$$FPR = \frac{FP}{FP + TN} \tag{5}$$

The area under the curve (AUC) quantifies the ability of the model to distinguish between the two classes (AD vs. healthy). A value of 1 refers to perfect classification, while a value of 0.5 suggests that the model performs no better than random guessing. AUC-ROC is particularly useful when comparing models with different thresholds for decision-making. Higher AUC values indicate better model discrimination between AD and non-AD cases. AUC-ROC is estimated via Eq. (6).

$$AUC = \int_0^1 TPRd(FPR) \tag{6}$$

10.6 Matthews correlation coefficient (MCC)

It is another metric that is valuable for evaluating binary classification models, especially when dealing with imbalanced datasets. Unlike accuracy, MCC takes into account all four confusion matrix elements (TP, TN, FP, FN) and provides a balanced metric of prediction performance. MCC ranges from -1 (perfect inverse prediction) to + 1 (perfect prediction), with 0 refering random predictions. MCC is particularly useful when both classes in the dataset are of similar importance, as it considers the trade-offs between false positives (FPs) and false negatives (FNs). The Eq. (7) represents the MCC.

$$MCC = \frac{(TP \times TN) - (FP \times FN)}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \tag{7}$$

10.7 Confusion matrix (CM)

It is not a metric by itself, but a valuable tool for visualizing a model's performance. It displays the counts of true positive, true negative, false positive, and false negative values in a matrix form, allowing for easy calculation of other measurements e.g., accuracy, F1-score, precision, recall. Table 5 illustrates statistical values of confusion matrix (CM). This matrix helps in understanding how well the model differentiates between two classes and where it makes confusions, which is essential to medical diagnoses where the consequences of false predictions can be significant.

10.8 Loss

It is often utilized to assess the performance of classification models that output probabilities rather than discrete labels. This metric quantifies the accuracy of the probabilistic

Table 5 Statistical values of confusion matrix

–	Predicted: negative	Predicted: positive
Actual: negative	*TN*	*FP*
Actual: positive	*FN*	*TP*

predictions, penalizing incorrect classifications more heavily when the model is confident about its wrong predictions. Equation (8) illustrates the log loss.

$$Loss = -\frac{1}{N}\sum_{i=1}^{N}\left[y_i log_2 \widehat{y}_i + (1-y_i)\, log_2\left(1-\widehat{y}_i\right)\right] \tag{8}$$

where y_i is the true label of the ith instance (0 or 1), while $\widehat{y}_i$ is the predicted label for the ith instance, and N is the total number of instances.

Lower loss values refer better model performance, with perfect predictions achieving a log loss of 0.

These evaluation metrics each offer distinct advantages according to the nature of the Alzheimer's disease classification problem. While accuracy provides a general sense of model performance, metrics like precision, F1-score, AUC-ROC, recall, and MCC give a more detailed understanding of model strengths and weaknesses, particularly in the case of imbalanced datasets. The use of confusion matrices and log loss further enhances model evaluation, providing deeper insights into the specific types of losses a model makes. Understanding these metrics is crucial for implementing a DL model that isn't only accurate, but also reliable and safe for use in clinical settings.

10.9 Error analysis in AI models for misclassification patterns

Error analysis is a crucial step in evaluating AI models, especially in high-stakes fields such as healthcare. By identifying common misclassification patterns, researchers and clinicians can improve model performance, enhance interpretability, and mitigate risks in clinical decision-making. There are several misclassification patterns often fall into several categories: false positives (Type I Errors), false negatives (Type II Errors), confusion between similar classes, bias towards majority class, and context-dependent errors. Moreover, several factors contribute to these misclassification errors: data imbalance, feature overlap, poor data quality, limited training data, and bias in training data. Misclassification in AI models can have serious consequences in healthcare: false positives: unnecessary interventions, false negatives: missed diagnoses, trust and adoption challenges, and ethical and legal concerns.

11 Potential bias in AI models and bias mitigation strategies

Bias in AI models for Alzheimer's Disease (AD) diagnosis presents significant challenges that can lead to disparities in healthcare outcomes. Ethical concerns, including fairness, transparency, and accountability, must be addressed to ensure AI-driven diagnostic tools benefit all patient populations equitably. By implementing robust bias mitigation strategies such as improving data diversity, auditing algorithms, and enhancing model transparency, we can reduce disparities and improve trust in AI-assisted healthcare.

Fairness in AI-driven AD diagnosis requires a collaborative effort between clinicians, data scientists, and policymakers to create models that are both accurate and ethically responsible. AI should serve as a decision-support tool that enhances, rather than replaces, human expertise. With ongoing monitoring and refinement, AI has the potential to revolutionize early AD detection, ensuring timely and precise diagnosis for all patients, regardless of their background. Also, AI tools should be designed to support equitable access to early AD screening and intervention, regardless of a patient's socioeconomic status or location.

12 Challenges, limitations, and comparisons of deep learning approaches for AD

This section presents many challenges in this field such as AD datasets are often limited in size and skewed towards certain classes, impacting model training. Also, models learned on particular datasets may struggle to predict across different populations and imaging protocols. In addition, deep learning models achieve high accuracy, their black-box nature hinders clinical adoption. Also, there are key challenges in AD detection such as early detection, imaging, heterogeneity of alzheimer's, lack of ground truth, longitudinal monitoring, and complexity of neurological data. Table 6 presents many challenges comparisons.

Also, a comprehensive comparison of various deep learning (DL) models is presented, their architectures, datasets, preprocessing techniques, evaluation metrics, and challenges. The aim is to highlight the strengths and weaknesses of different approaches. Table 7 shows some comparisons for models' architecture.

13 Computational cost and latency in clinical AI deployment

Deploying AI models in clinical settings involves computational costs and latency challenges. Deep learning models for AD detection, especially using MRI/PET, require high computational power and may not be feasible for real-time use. EEG- and speech-based detection methods are more suitable for real-time applications due to lower inference times (milliseconds to seconds). Real-time AD detection using EEG or speech analysis is feasible with lightweight AI models. Cloud-based solutions introduce network latency, while edge AI and hardware optimizations (e.g., model compression, FPGA acceleration) can improve feasibility. Balancing accuracy, cost, and computational efficiency is key to successful deployment in real-world clinical environments. Deep learning models for medical imaging

Table 6 Comparisons of challenges

Model type	Impact of deep learning model	Solutions
Limited dataset size	Transfer learning models mitigate this to an extent	Data augmentation, synthetic data generation
Imbalanced classes	CNNs may overfit to dominant classes	Use GANs or resampling techniques
Model interpretability	Black-box nature of CNNs and hybrids reduces clinical trust	Introduce explainable AI techniques (e.g., Grad-CAM)
Generalization	Poor cross-dataset performance of specialized models	Multi-domain training and testing
High computation costs	Complex models like hybrid architectures demand substantial resources	Optimize architectures, use cloud computing

Table 7 Comparisons for models' architecture

Model type	Advantages	Limitations
CNN	Extracts hierarchical spatial features, efficient for neuroimaging data	Requires large datasets to avoid overfitting; limited temporal analysis
Transfer learning	Leverages pre-trained models, reducing computation time and data requirements	Potential mismatch between pre-trained and medical domains
Hybrid models	Combines advantages of different architectures (e.g., CNN + RNN)	Increased complexity, computational overhead
Attention mechanisms	Focuses on clinically relevant regions, improving interpretability	May require large datasets for robust training of attention modules
GANs	Generates synthetic data for augmentation, addressing class imbalances	Can introduce artifacts affecting model robustness

(e.g., CNNs, transformers) require high computational power for inference, especially when processing high-resolution MRI, PET, or EEG scans. Large-scale models (e.g., deep neural networks with millions of parameters) demand powerful GPUs/TPUs, increasing deployment costs. Edge AI solutions (e.g., deploying models on local hospital servers or medical devices) must balance accuracy with computational efficiency. Inference time depends on hardware (CPU vs. GPU), model size, and input complexity.

14 New trends and suggestions in DL for Alzheimer disease detection

In this section, we discuss many directions aim to enhance accuracy, usability, and clinical impact in Alzheimer's research. There are many trends in this field are presented as follows:

1. Multi-modal data integration: Combining imaging (MRI, PET) with clinical, genetic, and behavioral data enhances model robustness and predictive power. Practical example: in Alzheimer's disease diagnosis, researchers combine MRI scans, PET imaging, genetic markers, and cognitive test scores to improve early detection. DL models integrate these diverse data types to enhance classification accuracy.
2. Transformers and attention mechanisms: Vision Transformers (ViT) and self-attention layers are increasingly used for improved feature extraction. Practical example: Vision Transformers (ViT) is used to analyze MRI and PET scans for early detection of AD. ViTs use self-attention mechanisms to capture long-range dependencies in brain imaging data, improving feature extraction and classification accuracy.
3. Explainable AI (XAI): Efforts to make models interpretable, aiding clinicians in understanding predictions and decisions. Practical example: in Alzheimer Disease detection from MRI scans, SHAP (SHapley Additive exPlanations) is used to highlight the most important pixels in an image that contribute to a model's decision, helping radiologists trust AI-driven predictions.

4. Federated learning: Enabling collaborative model training across institutions without sharing sensitive patient data. Practical example: a federated learning approach in Alzheimer Disease detection allows multiple hospitals to collaboratively train a deep learning model on MRI images without transferring patient data, preserving privacy while improving generalization.
5. Synthetic data generation: Techniques like GANs to augment datasets, addressing limited availability and class imbalance. Practical example: in rare disease diagnosis, GANs (Generative Adversarial Networks) generate synthetic MRI scans to balance underrepresented classes in datasets, improving model performance in detecting conditions with limited real data.

By comparing the mentioned five trends in terms of the models, datasets, techniques, and challenges, it becomes evident that the selection of approach depends on particular project goals, data availability, and computational resources.

Also, there are several suggestions in this field are introduced as follows:

1- Focus on early detection: Develop methods for early-stage diagnosis, such as MCI to Alzheimer's transition prediction.
2- Enhanced interpretability: Integrate XAI tools to build trust in clinical applications.
3- Ethical AI: Address biases in datasets to avoid disparities in diagnosis and treatment recommendations.
4- Cross-domain collaboration: Encourage partnerships between AI experts, neuroscientists, and clinicians to refine methodologies.
5- Real-world validation: Validate models on diverse, real-world datasets to ensure scalability and generalization.

15 Future directions in DL for Alzheimer disease detection

Several suggestions could be applied such as multi-modal learning to combine imaging data with genetic, clinical, and lifestyle information, also explainable AI (XAI) to develop interpretable models to gain clinicians' trust, Larger and diverse datasets to enhance model robustness across demographics, and real-time applications to translate DL models into clinical settings for real-time diagnostics. Also, there are many future works aim to make Alzheimer's disease classification more accurate, interpretable, and clinically viable. These works can be applied as follows:

1- Integration of multi-modal data: Future research can focus on fusing various data modalities like MRI, PET, clinical records, and genomic data for holistic disease modeling.
2- Advanced transfer learning: Expanding pre-trained models on domain-specific tasks to improve early detection accuracy and reduce the need for large labeled datasets.
3- Personalized models: Creating models tailored to individual patients by leveraging longitudinal data and personal health records.
4- Real-time diagnosis tools: Developing lightweight models deployable on edge devices for real-time, point-of-care screening.

5- Longitudinal prediction models: Exploring disease progression through temporal data analysis using RNNs, LSTMs, or Transformers.
6- Explainability improvements: Designing interpretable models to gain clinician trust by explaining predictions using heatmaps or feature attribution techniques.
7- Federated learning applications: Promoting decentralized training approaches to enhance collaboration while maintaining data privacy.
8- Synthetic data and augmentation: Using GANs and variational autoencoders to expand datasets for better model training.
9- Exploration of non-imaging biomarkers: Incorporating speech, handwriting, or gait analysis for non-invasive early diagnosis.
10- Bias mitigation: Ensuring models are unbiased by testing across diverse populations to generalize performance.

16 Benchmark comparison of DL models for AD detection

The following is a detailed benchmark comparison across various studies in the field of AD classification, focusing on the deep learning models used, datasets, and accuracy performance. In (Slimi et al. 2024), a hybrid DL model integrating DenseNet121 and Xception networks has demonstrated exceptional performance in AD classification. Utilizing the ADNI dataset comprising MRI images, this approach achieved an impressive overall accuracy of 99.85% through fivefold cross-validation. This model leverages the complementary strengths of DenseNet121 and Xception for feature extraction, providing a robust framework that enhances detection performance. Key innovations include using data augmentation techniques like SMOTE to balance datasets and enhance generalization. Furthermore, the model exhibits resilience against various image noise types, including Gaussian, Salt-and-Pepper, and Speckle, making it a robust solution for practical applications in Alzheimer's disease detection.

In (Turrisi et al. 2024), CNNs have been extensively used for Alzheimer's disease classification, leveraging various architectural designs to achieve robust performance. Utilizing the ADNI dataset, these models have consistently delivered accuracy rates ranging between 95 and 97%, with slight variations based on the specific architecture and the inclusion of data augmentation techniques. A key strength of CNN-based approaches is their focus on reproducibility, ensuring that experimental setups and results can be consistently validated. Moreover, the diversity in architectural choices—ranging from simple to highly complex designs—enables researchers to tailor models to the nuances of their datasets. The consistent application of cross-validation further reinforces the reliability of these models, making CNNs a foundational tool in Alzheimer's disease detection and classification.

In (Tong et al. 2024), multiple Instance Learning (MIL) has emerged as an approach for dementia classification, particularly in scenarios including weakly labeled or incomplete data. Using the OASIS dataset, which includes MRI scans and clinical data, MIL achieves accuracy rates between 85 and 90%, depending on the configuration. The model's strength lies in its ability to handle missing annotations and ambiguous information, which are common challenges in clinical environments. By evaluating sets of instances (e.g., slices of MRI scans or segments of clinical data) rather than requiring explicit labels for each instance, MIL effectively classifies dementia even with limited or incomplete information. This capability makes it a practical and efficient tool for real-world applications where fully labeled datasets are rare.

In (Morris et al. 2406), explainable AI (XAI) has been increasingly integrated into CNNs for AD diagnosis, focusing on enhancing the interpretability of AI-driven decisions. Using the ADNI dataset of MRI images, these models achieve accuracy levels of 96–98%, comparable to traditional CNNs. However, the distinguishing contribution of XAI-based models lies in their use of methods such as saliency maps and attention mechanisms to provide visual or conceptual explanations for their predictions. This capability helps build trust among clinicians by offering insights into how the model identifies Alzheimer's-related patterns, such as brain atrophy. By addressing the "black box" nature of DL, XAI-driven models promote the adoption of AI in clinical settings, fostering collaboration between AI systems and medical practitioners through transparent and interpretable decision-making processes.

In (Khan et al. 2022), the study explored hybrid DL models—comprising CNN, Bidirectional LSTM, and Stacked Deep Dense Neural Network (SDDNN)—for early Alzheimer's disease (AD) detection through text classification of clinical transcripts. Leveraging the DementiaBank dataset, the models were trained and evaluated using both randomly initialized weights and pre-trained GloVe embeddings. Extensive hyperparameter tuning was performed through GridSearch method. Among these models, the SDDNN mixed with GloVe embeddings reached the highest accuracy of 93.31% and outperformed others in metrics such as AUC and recall. The results underscore the promise of automated approaches in assisting clinicians with early AD diagnosis, while emphasizing the need for further research to enhance performance on larger datasets.

In (Ramani et al. 2024), three-dimensional Convolutional Neural Networks (3D CNNs) have proven highly effective for the early detection of AD by leveraging 3D volumetric data from MRI scans. Utilizing the ADNI dataset, these models achieve accuracy rates of 93% to 97%, depending on their specific configurations and pre-processing approaches. The primary strength of 3D CNNs is their ability to capture spatial dependencies across the three-dimensional structure of the brain, enabling the detection of subtle changes in brain regions associated with early Alzheimer's. Unlike traditional 2D CNNs, which analyze slices of imaging data independently, 3D CNNs analyze the complete volumetric context, offering improved sensitivity to early structural alterations. This capability makes 3D CNNs particularly valuable for diagnosing Alzheimer's disease at its earliest stages, facilitating timely interventions.

In (Shastry 2024), a custom CNN has been developed using the ADNI dataset, which includes 10,000 MRI images categorized into three classes: Non-demented, MCI, and AD. The methodology involved incorporating regularization methods, e.g. dropout, and employing data augmentation techniques to improve the model's robustness and generalizability. These strategies significantly enhanced classification performance, achieving high accuracy while mitigating overfitting by expanding the effective training dataset with synthetic variations. However, the custom CNN faces challenges, including limited interpretability, which restricts its ability to provide explanations for its predictions—a critical factor for clinical adoption. Additionally, the model is computationally expensive to train, requiring substantial computational power and large memory, which may limit its scalability in resource-constrained settings. Despite these limitations, the approach demonstrates a powerful framework for AD detection.

In (Nanthini et al. 2024), the Hybrid Deep Belief Network (DBN) approach integrates imaging and non-imaging data using a multi-task learning framework, applied to a combined dataset of ADNI and OASIS with 15,000 images. The data is categorized into 3 classes: Cognitive Normal, MCI, and AD. The methodology leverages the representational power of Deep Belief Networks (DBNs) to process multi-modal data, improving the ability to extract complex patterns and dependencies. This approach excels in robust multi-modal learning and provides effective predictions of disease progression, particularly by synthesizing diverse data types like MRI scans and clinical metrics. However, the integration of multi-modal data imposes high computational demands, requiring substantial resources for training and deployment. Despite these challenges, the Hybrid DBN method offers a promising avenue for improving diagnostic accuracy and understanding Alzheimer's disease progression.

Hussain et al. (2020) introduced a 12-layer CNN model to classify two categories (Alzheimer/healthy) on MRI data from the OASIS dataset, achieving 97.75% accuracy and 97.50% F1-score, surpassing pre-trained architectures like InceptionV3 and VGG. Erdogmus and Cui et al. (2021) utilized adaptive logistic regression with particle swarm optimization (PSO) on the ADNI dataset, reporting accuracy values of 96.27%, 84.81%, and 76.13% for different binary classifications of Alzheimer's subtypes. Erdogmus and Kabakus (Erdogmus and Kabakus 2023) proposed a 12-layer CNN optimized using 12 hyperparameters for the DARWIN dataset, which involved converting 1D data to 2D, yielding a classification accuracy of 90.4%.

Sun et al. (2021) enhanced ResNet50 with spatial transformer networks and attention mechanisms, achieving 97.1% accuracy. Manimurugan (2020) fine-tuned VGG19 on OASIS data to achieve 95.82% accuracy, while Sharma et al. (2022) employed a VGG16-based model with ANN for Alzheimer's classification, obtaining 90.4% accuracy for four-class classification. Savas (2022) identified EfficientNetB0 as the top-performing model in a comparative analysis with a 92.98% accuracy rate. Lahmiri (2023) combined CNN-based feature extraction with KNN classification optimized using Bayesian optimization, achieving 94.96% accuracy on OASIS data.

There are several common challenges were encountered across the mentioned studies on Alzheimer's disease classification using deep learning techniques: one of the primary issues is data availability and quality. Many studies rely on publicly available datasets such as ADNI, OASIS, and DementiaBank, which, while valuable, may not always be representative of diverse populations. Additionally, limited data can lead to overfitting, especially in complex models requiring extensive training.

Another significant challenge is class imbalance. Some datasets contain an uneven distribution of classes, with fewer samples for early-stage Alzheimer's or Mild Cognitive Impairment (MCI). This imbalance can bias models toward majority classes, reducing performance for underrepresented groups. Researchers often apply techniques such as oversampling, undersampling, or weighted loss functions to address this issue, but it remains a persistent limitation.

Interpretability and explainability also pose critical concerns. Deep learning models, particularly CNNs, often function as "black boxes," making it difficult for clinicians to trust and interpret their decisions. Although explainable AI (XAI) methods like saliency maps and attention mechanisms have been introduced, their effectiveness in real-world clinical decision-making remains an ongoing challenge. Without clear interpretability, AI models may struggle to gain acceptance in medical practice.

The computational complexity of deep learning models is another hurdle. Advanced architectures, such as 3D CNNs and hybrid deep learning models, demand substantial computational resources, making them difficult to implement in clinical settings with limited infrastructure. Training deep models on large MRI datasets requires powerful GPUs and extended processing time, limiting accessibility for many research institutions.

Generalizability and reproducibility are also key concerns. While cross-validation techniques enhance model reliability, models trained on one dataset (e.g., ADNI) may not generalize well to others (e.g., OASIS or DARWIN) due to variations in imaging protocols, demographics, and clinical criteria. Ensuring that models remain robust across different data sources is a significant challenge that researchers must address.

Handling noisy and incomplete data presents further difficulties. Clinical datasets often contain missing or weakly labeled information, requiring techniques like Multiple Instance Learning (MIL) to compensate. Noisy data, arising from imaging artifacts or inconsistent labeling, can reduce model performance and lead to unreliable predictions.

Another challenge is feature extraction and data representation. The effectiveness of deep learning depends on the quality of extracted features, but determining the most relevant features for Alzheimer's detection remains complex. Multi-modal learning approaches that integrate MRI scans, clinical records, and cognitive assessments can improve accuracy but also introduce additional complexity in feature fusion and model design.

Overfitting and regularization are also recurrent issues. Many studies applied techniques such as dropout, data augmentation, and hyperparameter tuning to mitigate overfitting. However, ensuring that models remain robust across different datasets is still a challenge. Some models achieve high accuracy on training data but fail to generalize well on unseen test data.

Finally, clinical adoption and validation remain major barriers. Despite the high reported accuracy of many models, deep learning approaches require extensive validation before clinical deployment. Regulatory challenges, ethical concerns, and the need for seamless integration into existing medical workflows hinder real-world implementation. Without addressing these concerns, AI-based models may struggle to transition from research settings to practical healthcare applications.

Table 8 presents a comparison for several studies in AD detection in terms of the methodology, dataset, AI models, accuracy, and key contributions. The used criteria is the accuracy metric for the same application of Alzheimer disease (AD). All studies include AI models that used for AD detection applications, these models are compared in terms of the accuracy to evaluate their performance.

The architecture of a deep learning model is selected based on some factors for instance number of parameters. So, there are many considerations to choose a model. For example, the Inception-V3 model has a higher accuracy than VGG-19 and it needs less size on the RAM memory than VGG-19, due to the small number of parameters for the Inception-V3. Therefore, it requires has computational complexity (more mathematical computations/ operations), which lead to consuming more training time.

Table 8 A comparison for several studies in AD detection

Study	Methodology	Dataset	AI models	Accuracy	Key contributions
Slimi et al. (2024)	Hybrid Deep Learning Model (DenseNet121 & Xception)	ADNI (MRI images)	DenseNet121, Xception (hybrid)	99.85%	Feature fusion, noise robustness, high accuracy, SMOTE for data augmentation
Turrisi et al. (2024)	CNN-Based Model for AD Classification	ADNI	Various CNN architectures	95–97%	Reproducibility, consistent evaluation, performance with data augmentation
Tong et al. (2024)	Multiple Instance Learning for Dementia Classification	OASIS (MRI + clinical)	Multiple Instance Learning (MIL)	85–90%	Handling weakly labeled data, dealing with incomplete clinical data
Morris et al. (2406)	Explainable AI (XAI) for AD Diagnosis	ADNI	CNN + XAI techniques (e.g., saliency maps, attention)	96–98%	Transparency, trust-building in clinical settings, interpretable decision-making
Khan et al. (2022)	Implementing three deep learning models with a significant hyperparameter optimization	DementiaBank	Hybrid models (CNN, Bidirectional LSTM), SDDNN	93.31%	Achieving significant hyperparameter optimization via GridSearch
Ramani et al. (2024)	3D CNN for Early AD Detection	ADNI (3D MRI volumes)	3D CNN (volumetric data)	93–97%	3D spatial dependencies for early detection, capturing subtle brain changes in volumetric MRI data
Shastry (2024)	Custom CNN + data augmentation	ADNI	CNN	–	High accuracy, reduced overfitting
Nanthini et al. (2024)	Multi-task learning	ADNI + OASIS	DBN + Multi-task	–	Effective progression prediction
Hussain et al. (2020)	Applying three deep learning: models for binary classification (Alzheimer/healthy) on MRI	OASIS	CNN, InceptionV3, and VGG	97.75	Apply CNN for both binary and multi classification
Cui et al. (2021)	Implementing adaptive logistic regression with particle swarm optimization (PSO)	ADNI	Developing PSO as an optimization algorithm	96.27	High accuracy, reduced overfitting
Erdogmus and Kabakus (2023)	Building 12-layers CNN optimized using twelve hyperparameters	DARWIN	CNN	90.4	Various processing on huge dataset

Table 8 (continued)

Study	Methodology	Dataset	AI models	Accuracy	Key contributions
Sun et al. (2021)	Enhancing ResNet50 with spatial transformer networks and attention mechanisms	ADNI	Hybrid of Transfer Learning	97.10	A new ResNet model incorporating the Mish activation function
Manimurugan (2020)	Applying transfer learning for multi-classification	OASIS	VGG19	94.82	Fine tuning for the hyperparameters
Sharma et al. (2022)	Employing a VGG16-based model with ANN	Kaggle	VGG16	90.4	Features from MRI scan images are extracted using the VGG16 model
Savaş (2022)	Implementing EfficientNetB0 as the top-performing model	ADNI	Transfer Learning	92.98	Apply several processing techniques on the dataset
Lahmiri (2023)	CNN-based feature extraction with KNN classification	OASIS	Developing CNN with an optimization algorithm	94.96	Applying processing with both ML and DL models

17 Conclusions

This paper examines the latest advancements in DL and ML models for AD classification. It also explores various applications of AI in AD research, including datasets, preprocessing methods, challenges, and notable recent studies in the field. Furthermore, the paper discusses medical imaging modalities, risk factors associated with AD, the disease's progression stages, and key metrics used to evaluate the performance of AI models. Additionally, it provides comparative analyses of different DL approaches, highlights their limitations, identifies emerging trends, and offers recommendations and future directions for this rapidly evolving domain.

In conclusion, deep learning has emerged as a transformative tool in the detection and classification of AD, offering potential for early diagnosis and personalized treatment. While advancements in multi-modal integration, transfer learning, and model interpretability have been made, challenges such as data scarcity, computational complexity, and model generalization remain.

Deep learning has proven to be a game-changer in Alzheimer's disease classification, offering unprecedented accuracy and efficiency. While challenges remain, continued advancements in model architectures, data availability, and interpretability promise to cover the gap between research and clinical practice. This paper presents a review to underscore the potential of DL models as critical tools in the fight against AD.

In the context of evaluation metrics for AD detection, it is essential to use a combination of the evaluation metrics to ensure reliable and accurate results. Proper evaluation allows for the optimization of DL models, making them more effective in clinical practice and improving early detection of AD.

Future work should focus on refining these models, ensuring clinical applicability, and promoting collaborations to enhance diagnostic accuracy and expand access to AD care, ultimately enhancing patient outcomes and advancing healthcare solutions.

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Declarations

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